Hierarchical models to predict predation in complex trophic communities

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Trophic interactions make the backbone of ecological communities. From herbivory to predation, these interactions of varying strength shape ecosystem and their related functions. There is a pressing need to be able to predict interaction strength at the scale of whole communities. While functional responses focus at the pairwise interaction level, we question here if the same underlying mechanisms could be leveraged to make predictions at the scale of the community. In this study, we develop 4 hierarchical models of increasing complexity based on the type I and the type II functional responses to predict biomass fluxes between consumers and resources at the community level. The main objective is to determine if simple mechanism such as mass action (type I) and the saturation of fluxes (type II) are sufficient to predict biomass fluxes. To do so, we collected 19 Ecopath networks encompassing 1380 interactions across marine, aquatic and terrestrial ecosystems. While we can not reject the mechanism of saturating fluxes (type II), our results suggest that a mass action mechanism is sufficient and more robust, in the context of our data, to predict biomass fluxes between consumers and resources at the community level. We also recovered the expected allometric relationships between consumers body mass and their space clearance rate parameters within the mass action model, further solidifying our findings. Further modelisation is needed, specifically a trait-matching model with more data, to develop a model generalizable to more taxons and types of trophic interactions.

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Trophic interactions make the backbone of ecological communities (Bellingeri & Bodini 2016; DeLong 2021; Wootton & Emmerson 2005), linking every component through nutrient and energy flows (Lindeman 1942). Consumer-resource interactions of varying strength play a crucial role in shaping important ecosystem functions and processes (Curtsdotter *et al.* 2019; Duffy 2002; Koltz *et al.* 2018; Montoya *et al.* 2003; Wilmers *et al.* 2012) such as disease dynamics, invasive species and biogeochemical cycles (Estes *et al.* 2011). Identifying species interactions and quantifying their strength is necessary to understand community composition (Paine 1992), stability (Bascompte *et al.* 2005; de Ruiter *et al.* 1995; Neutel 2002; Nilsson & McCann 2016) and the resulting associated ecosystem services (Montoya *et al.* 2003). Global changes have a diversity of impacts on species themselves, but also how they interact and the resulting assemblages and functioning (Koltz *et al.* 2018; Lurgi *et al.* 2012). Empirical observation of these interactions and the evaluation of their strength is however challenging, if not unfeasable due to their sheer amount, sampling biases and the emergence of novel interactions between species that never co-occurred (Aguiar *et al.* 2019; Jordano 2016; Wootton & Emmerson 2005). The study of ecological networks has therefore recently moved to the development of predictive tools to overcome the limitations associated with empirical sampling of trophic interactions (Poisot *et al.* 2016). Different models are derived from food web theory, such as the niche model (Williams & Martinez 2000), the nested-hierarchy model (Cattin *et al.* 2004), trait-matching model (Bartomeus *et al.* 2016) or physical constraints (Portalier *et al.* 2019). More recently, Strydom *et al.* (2021) developed a general framework to predict interaction probabilities between species in a community context. These methods are however mostly qualitative, i.e. they are meant to predict if there is an interaction or not, and there is a pressing need to expand them to quantitative interactions, where the strength of interaction between any two organisms is described.

There are many definitions of interaction strength (Berlow *et al.* 2004), and among them biomass flows can be particularly suited for the development of predictive models. Biomass flows between consumers and resources can be derived from functional traits, with the promise of a standardized measure independent of taxa involved (Berlow *et al.* 2004). Quantitative predictive models of biomass flows might also offer potential support to the study of biodiversity and ecosystem functioning relationship, as it seeks to depict all energy flows within communities (Barnes *et al.* 2018). Biomass flows are defined as a quantity of resource biomass that is consumed by a consumer per unit area and unit of time, reporting how energy circulates through the food web (Lindeman 1942). Interaction strengths between consumers and resources is closely related to predation rate in a food web model, namely the functional responses](Rall *et al.* 2012). Functional response theory aims at describing the change in feeding rate of a consumer, or how many prey or resource a consumer can predate per

unit area over time, relative to prey or resource availability (Holling 1959a, b; Solomon 1949). Where the functional response describes the actual number of individual prey or resource predated by a consumer per unit area over time, biomass flows describes the amount of biomass flowing from a prey or resource to a consumer per unit area over time. Since both quantities report the rate of the same underlying process that is predation, we anchored the development of our modeling approach in functional response theory.

A fundamental principle underlying the majority of functional responses is the mass action, stating that predator and prey move randomly in an area and encounter at a rate proportional to the product of their abundance and their velocity. Here we will focus on the two most commonly used types of functional responses. The type I functional response describes mass action only, where the amount of resource consumed depends on the resource and consumer densities, and the rate (formerly reffered to as attack rate and hereafter space clearance rate (DeLong 2021)) at which a consumer searches its home range for resources, leading to random encounters followed by detection, attack and consumption. The type II functional response describes the same process as the type I, but with the addition that consumption saturates with prey availability because of the time required to process prey. These processes are influenced by a multitude of functional traits and environmental conditions (Beardsell *et al.* 2021, 2022). The space clearance rate parameter varies with consumers and resources intrinsic characteristics such as absolute and relative velocities, perception of the environment and cognitive abilities (DeLong 2021). It is also influenced by externalities such as habitat complexity (Barrios-O'Neill *et al.* 2016; Hall *et al.* 2000; Manatunge *et al.* 2000) and the presence of refugia (Hoddle 2003) impeding detection probability. The dimensionality of an interaction, whether it takes place in a two or three dimensional plane, also impacts encounter rates and detection probability (Pawar *et al.* 2012, 2019). For instance, a predatory bird searching for preys from above or fishes grazing the benthic surface are examples of 2D planes while aerial insectivorous birds pursuing flying insects or fishes chasing preys in the water column would be in a 3D plane. Functional response theory therefore makes a good start to develop predictive models of quantitative interactions since widely measured functional traits can be combined with consumer and prey density to predict biomass flows.

The space clearance rate, in units of space per time, reports the amount of area search by the predator in a time interval. As with many biological processes, the space clearance rate was shown to vary with consumers body mass. Indeed, multiple studies, summarized in the FoRAGE database (Uiterwaal *et al.* 2022), showed an allometric scaling between the space clearance rate and the body mass of consumers. This observation is coherent with (Hirt *et al.* 2017, 2020) who documented that the maximum velocity of an organism scales with

its body mass. Handling time is also expected to follow the same scaling relationship (DeLong 2021), although recent studies are now suggesting that consumers handling time would not be as important at natural resources densities (Beardsell *et al.* 2021; Chan *et al.* 2017; Coblentz *et al.* 2023; Preston *et al.* 2018). Ultimately, an accurate evaluation of these parameters is necessary as they are widely used in theoretical models (Brose *et al.* 2006; Brose *et al.* 2008; Pawar *et al.* 2012). Restraining the parameter space could greatly bonify studies of species coexistence, community stability and trophic regulation. It could also help the parameterization of global ecosystem models used for biodiversity scenarios (Harfoot *et al.* 2014}). Following (Yodzis & Innes 1992), there are a few meta-analyses looking at the scaling of feeding rates with body-mass (Rall *et al.* 2012). These are however concentrated on a re-analysis of univariate functional responses, mostly from experimental studies, and there is currently no study aiming at reconstructing quantitative interaction networks.

Our main objective is to test if we could model quantitative interactions in complex food webs with easily accessible information such as abundance and body mass. We compared statistical models derived from functional response theory, including type I and type II forms. We also investigate if allometric relationships between space clearance rate, handling time and consumer body mass can contribute to the prediction of biomass flows in diverse food webs. To do so, we developed four hierarchical models of pairwise quantitative interactions of increasing complexity and evaluated them with empirical data of biomass flows between consumers and resources. We curated Ecopath networks (Christensen *et al.* 2005) describing marine, aquatic and terrestrial food webs and extracted 1380 pairwise trophic interactions between consumers and resources. Each model represents its own hypothesis and their comparison allow the evaluation of different mechanisms driving interactions.

Material and methods

A litterature search was performed to gather empirically sampled trophic networks with measurements of biomass flows from resources to consumers, along with population densities. Food web data are notoriously heterogeneous (Mestre *et al.* 2022) and we therefore relied on Ecopath models (Christensen *et al.* 2005) to normalize data collection. Ecopath is a modelling software that relies on a mass-balance premiss and a multitude of parameters (production/biomass ratio, consumption/biomass ratio, diet compositions etc.) to provide a realistic static image of the flows of biomass between resources and consumers in an ecosystem (Christensen *et al.* 2005). It was shown the Ecopath methodology reduces biases in comparative analyses

among food webs (Brimacombe, in review). We therefore collected openly available models from Ecobase (Colléter *et al.* 2013) and worked with a subset of the models present in (Jacquet *et al.* 2016). Taxonomic resolution of Ecopath models is highly variable and it is not uncommon that species are lumped into functional groups. The models were therefore selected based on the taxonomic resolution with the criteria that most nodes must be resolved at the species level, or that species within functional groups were taxonomically resolved. We thus curated 19 Ecopath models (detailed list in supplementary material) from aquatic, marine and terrestrial environments (fig. 1), and trophic guilds spanning from mammal carnivores and herbivores, birds, pelagic carnnivorous and herbivorous fish, demersal carnivorous and herbivorous fish and invertebrates.

[Figure 1 about here.]

Species identity in the Ecopath models were validated with the original publication and estimates of adult mean body mass was retrived from literature. Body mass for terrestrial species came from different sources such as the GATEWAy database (Brose 2018), the original publication for which the Ecopath model was developped or grey literature. Body mass for marine and freshwater species came from Fishbase (Froese *et al.* 2023) and Sealifebase (Palomares *et al.* 2023). When only body length was available, Fishbase and Sealifebase calculated weight from body length using a bayesian length-weight model. The midpoint between the minimum and maximum measurements was taken when a mean was not available. Density (number of individuals per unit surface, N/km^2) was computed by dividing total biomass per unit surface B (metric tons/km^2) by mean body mass M (metric tons). The 19 Ecopath models were aggregated in an edge list with quantitative biomass flows (metric tons/km^2/year), for a total of 1380 interactions between 154 consumer species and 153 resource species.

Model description

We developed four different hierarchical models of increasing complexity based on the type I and type II functional responses, and a null model. Models were then parametrized with Hamiltonian Monte Carlo (Monnahan *et al.* 2017), compared and ranked from better to worst based on their prediction accuracy. The functional form of these relationships are presented below.

Model 0 - Null model

The null model acts as a basis of comparison with the other models and does not portray any ecological realism of a trophic interaction. Model 0 hypothesizes that the flow of biomass F_{ij} from a prey i to its predator j is constant with average K, irrespective of predator identity.

$$F_{ij} = K$$

Model 1 - General mass action

Model 1 implements a general version of the type I functional response and thereby focuses on the law of mass action. The flow of biomass F_{ij} depends on the available prey biomass B_i and the abundance of the predator N_j where they encounter each other at a given the α space clearance rate of the consumer N_j. Explicitely, only the consumer's mobility is considered as it searches for resources. For the purpose of comparison, model 1 hypothesizes that all consumers have the same space clearance rate α.

$$F_{ij} = \alpha B_i N_j$$

Model 2 - Consumer-specific mass action

Model 2 follows the previous one with the hypothesis that space clearance parameter α_j is a specific parameter for each consumer species j.

$$F_{ij} = \alpha_j B_i N_j$$

Model 3 - Single-species saturating model

Model 3 keeps the assumptions of model 2 with the addition of handling time, thereby representing a single species Type II functional response. The space clearance rate parameter α_j and the handling time h_j are both consumer-specific:

$$F_{ij} = \frac{\alpha_j B_i N_j}{1 + h_j \alpha_j B_i N_j}$$

Model 4 - Multi-species saturating model

Building on model 3, model 4 implements a multi-species version of Type II functional response (Smout *et al.* 2010). Parameters are the same as the ones used in model 3, the difference is found at the denominator where handling is aggregated over all species in the food web :

$$F_{ij} = \frac{\alpha_j B_i N_j}{1 + h_j \alpha_j \sum_{i=1}^{nprey} B_i N_j}$$

Model name	Model equation	Number of parameters
Null model	$F_{ij} = K$	2
General mass action	$F_{ij} = \alpha B_i N_j$	2
Predator specific mass action	$F_{ij} = \alpha_j B_i N_j$	157
Single species saturating model	$F_{ij} = \frac{\alpha_j B_i N_j}{1+h_j \alpha_j B_i N_j}$	313
Multi species saturating model	$F_{ij} = \frac{\alpha_j B_i N_j}{1+h_j \alpha_j \sum_{i=1}^{nprey} B_i N_j}$	313

F_{ij} represents the flux of biomass (metric tons/km^2/year), K is a parameter representing mean biomass flow (metric tons/km^2/year), α is the general space clearance rate (km^2/consumer/year), α_j is the consumer-specific space clearance rate (km^2/consumer/year), B_i is the prey biomass (metric tons/km^2), N_j is the consumer abundance (individual/km^2) and h_j is the consumer handling time (metric tons of resource biomass/km^2/year). {#tbl:table_model}

Model evaluation

All biomass fluxes and body-mass were log-transformed. Biomass fluxes were modelled with a hierarchical bayesian approach where F_{ij} follows, after being log-transformed, a normal distribution with mean μ and standard deviation σ, where the mean corresponds to the five different models described above respectively. Species-specific parameters α_j and h_j were considered as random effects following normal distributions. As we lacked prior knowledge on potentiel distribution for the parameters, we opted for generic weakly informative priors. The mean and standard deviation for α and h_j were 0 and 1 respectively and were the same for all models. The analysis was performed with Stan version 2.32.2 (Stan Development Team 2024) via the package rstan in R version 4.3.3 (R Core Team 2024).

After fitting the data to each model respectively, we used the LOO package (Vehtari *et al.* 2024) to compute the expected log pointwise predictive density (ELPD) to perform a leave-one-out (LOO) cross validation based on (Vehtari *et al.* 2017) to estimate prediction accuracy. We computed Pareto diagnostic values per model to evaluate reliability. Models were ranked by predictive accuracy based on their respective ELPD score. We also compared the ELPD score between models to evaluate prediction accuracy and their bayesian R^2 (Gelman *et al.* 2019). Finally, we tested with linear regression allometric relationships of the space clearance rate parameter α and handling time h_j with consumer body mass.

Code and data availibility

All the R scripts for model evaluation and figures are available on Github at https://github.com/BenMerSci/master.

Results

All models (with the exception of the null hypothesis) had good fit to the data, with R^2 ranging from 0.697 (model 1 - details at tbl. **??** and R^2 posterior distributions in fig. **??**) to 0.835 (model 3). The model ranking based on the predictive accuracy with LOO, where higher values are better, revealed that the best performing model is the single species saturating model (model 3) with an ELPD score of -2452.8, almost equal to the multi species saturating model (model 4) and the consumer specific mass action model (model 2) with ELPD scores of -2713.5 and -2749.4 respectively. The the null model (model 0) and the general mass action model

(model 1) followed with ELPD score of -3529.1 and -4056.1, respectively. Model ranking reveals that, with the exception of model 4, every hypothesis that is added to the model contribute significantly to ELPD and R^2. The most significant increases is observed between model 1 and model 2, indicating that, once we account for mass action (model 1), the most important element is predator specific consumption rate (model 2). Adding saturation only marginaly improves the fit (model 3), while accounting for multi-species saturation has no significant impact (model 4).

It is to be noted that both model 3 and model 4 had higher Pareto K diagnostic values (> 0.7), which might indicate non-robust model or highly influential observations (Vehtari *et al.* 2017).

Model	Expected log pointwise predictive density (ELPD)	ΔELPD	Standard error difference	Effective number of parameters (p-LOO)	Number of parameters	mean R^2
model 3	-2455.4	0.0	0.0	202.5	313	0.833
model 4	-2717.7	-262.3	28.8	149.5	313	0.795
model 2	-2751.3	-295.9	30.2	142.2	157	0.803
model 0	-3529.1	-1073.7	35.7	2.4	2	0.000
model 1	-4056.5	-1601.1	37.3	2.1	2	0.696

Model ranking based on ELPD with LOO. Models are ranked from best to worst based on their predictive accuracy with the LOO package. {#tbl:table_ranking}

Model comparison suggests that mass action only catches most of the variability in biomass fluxes, but the graphical representation of predicted biomass flows against observed biomass flows reveals a very significant bias that is solved with species-specific space clearance rates (figure fig. 2). Since the bias in predictions disappears in all three species-specific models and that model 1 ranked below our null model, it was dismissed. Model ranking and the R^2 suggest the single species saturating model (model 3) is better than model 2 and 4, but the difference is hardly significant both from statistics (Table tbl. **??**) and graphical representation (figure fig. 2). The consumer specific mass action model (model 2) and the multi-species saturating model (model 4)

are the closest together regarding their ELPD score. Furthermore, the added multi-species aspect of model 4, relative to the single-species model 3, does not improve the fit, neither from statistics nor from graphical representation, therefore dismissing model 4. The *extreme* Pareto K values of model 3 and 4 would also suggest that both models might not be suited to make prediction as is (Vehtari *et al.* 2017).

[Figure 2 about here.]

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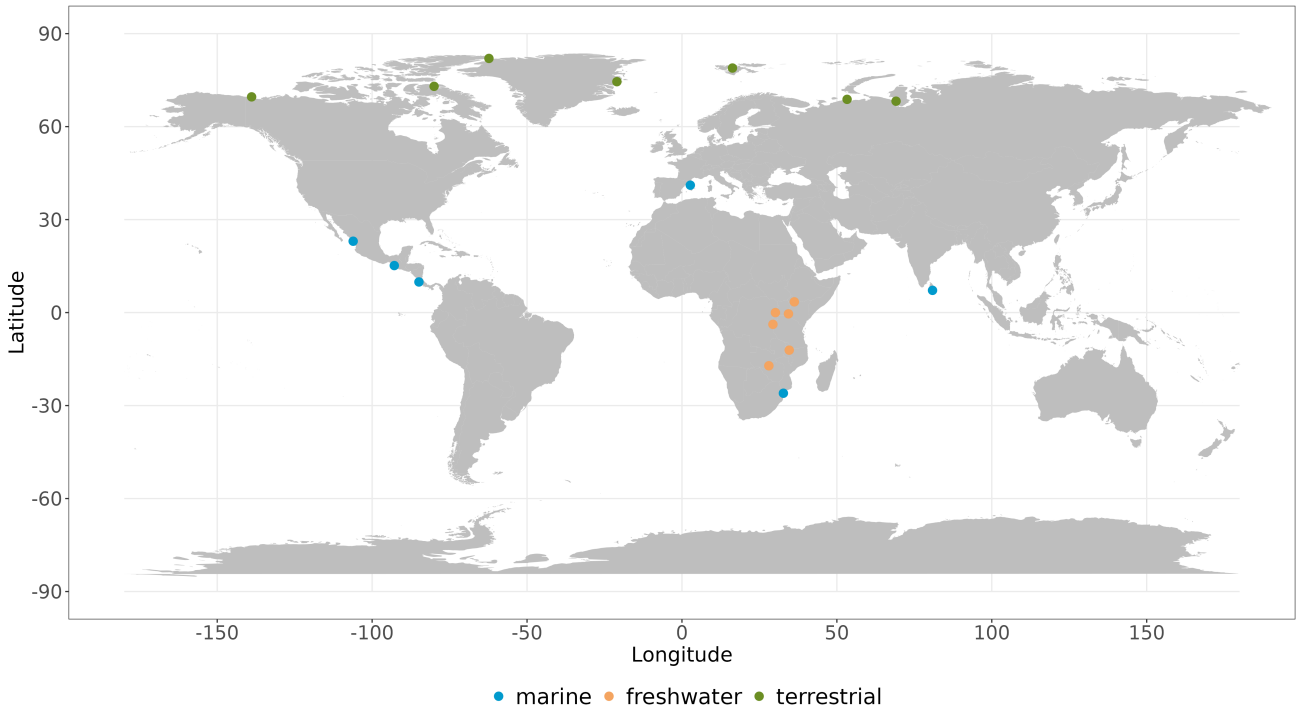

Figure 1: Map of Ecopath models locations used in the present study.

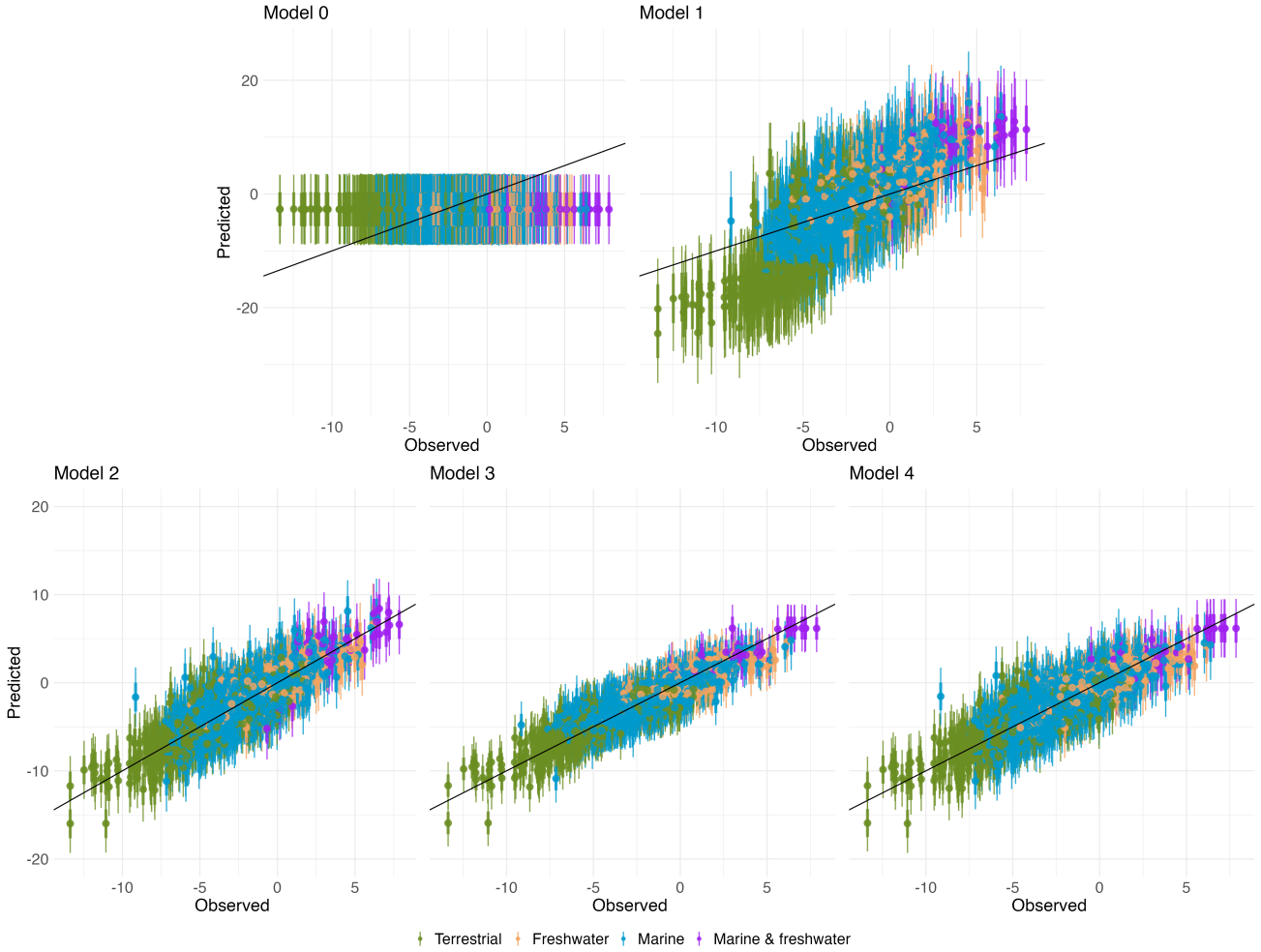

Figure 2: Models prediction of biomass fluxes compared to the observed biomass fluxes (log-scale) (metric tons/km^2/year). The black line represents a one for one line. Points represent the mean predicted value and thick and fine error bars represent the 0.66 and 0.95 quantile distribution respectively. 'Marine and freshwater' points represent biomass fluxes from consumers that are present both in marine and freshwater ecosystems such as Zooplankton and Zoobenthos.